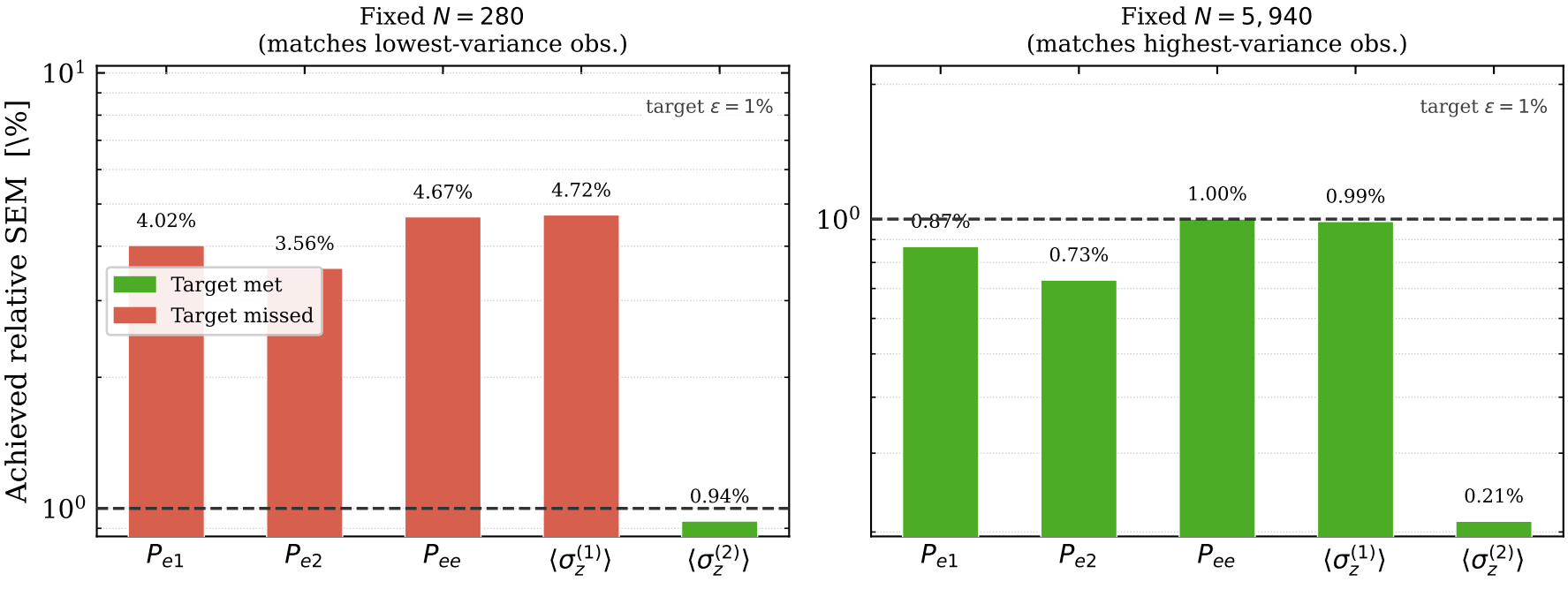
No single fixed N works for every observable, even at $d = 4$

Left: N tuned to $\langle\sigma_z^{(2)}\rangle$ starves the others. Right: N tuned to $\langle\sigma_z^{(1)}\rangle$ oversamples $\langle\sigma_z^{(2)}\rangle$ by 21×.